

XRPL Validator — Daily Engineering Report • 2026-09-29

Generated 2026-09-30 00:21 EDT. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **3 commits** (xrpl-validator 3)
- **2 engine findings fixed** (F407–F408)
- Live shadow: 12,404/12,404 ledgers fully matched, 1,807,964 transactions, 0 engine receipts in the current soak
- Amendment **PermissionDelegationV1_1** held majority (since 2026-09-24 21:25 UTC); activation due ~2026-10-08 21:25 UTC
- Amendment **fixBatchV1_2** held majority (since 2026-09-25 14:12 UTC); activation due ~2026-10-09 14:12 UTC
- Amendment **BatchV1_1** held majority (since 2026-09-25 14:46 UTC); activation due ~2026-10-09 14:46 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	12,404
Ledgers fully matched	12,404
Transactions applied natively	1,807,964
Result mismatches	0
Byte mismatches	0
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	0
Live engine build running since	2026-09-29 10:57 EDT
Reference rippled	3.4.1, full, validated #107,329,549

Engine findings fixed

- **fix(xrpl-ledger): F407 — a stored QualityIn of QUALITY_ONE is no interest; the default line is deleted** (e1140ef, xrpl-validator) Mainnet #107311328 59262A2D67A7: a TrustSet (limit 0, tfSetNoRipple + tfClearFreeze) on a CNY line whose holder's side stores HighQualityIn = 1,000,000,000 (QUALITY_ONE) and otherwise has no interest: limit 0, balance 0, NoRipple set on a holder without lsfDefaultRipple. rippled's SetTrust reads the line's stored qualities and zeroes QUALITY_ONE before the reserve decision (TrustSet.cpp `if (QUALITY\_ONE == ...`

- **fix(xrpl-ledger): F408 — an AMM deposit mints the tokens its mode computes; LPTokenOut is never the mint** (1fd1b9b , xrpl-validator) Mainnet #107314214 5FC90948A0CD: a tfSingleAsset AMMDeposit of 1,021,192 drops into an XRP/FUZZY pool with LPTokenOut 56,639.2869999. In tfSingleAsset mode LPTokenOut is an optional MINIMUM: singleDeposit mints what the Amount buys (lpTokensOut, adjustLPTokensOut, adjustAssetInByTokens) and hands the field to deposit() as lpTokensDepositMin, which refuses tecAMM_FAILED when lpTokensDepositActual < ...

Engineering changes

- **fix(xrpl-node): the era sentinel walks the AccountRoot's fields; an unverified ledger re-hydrates the native shadow** (50a05ae , xrpl-validator) Mainnet #107314806 (soak #35, 08:26 EDT 2026-09-29): the era sentinel read the Sequence of the ledger's first seq-bearing transaction's sender (Payment C4F7C77C779F, Sequence 62,037,183) out of its AccountRoot with a byte search for 0x24. The account's Flags are 0x24000000 (lsfDisallowIncomingTrustline | lsfDisallowIncomingNFTokenOffer), so the search hit the Flags' first byte and read Sequence 36: a false ...

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Soak #35: two engine receipts, both fixed; ended by the deployment below.** Soak #35 of build 0145757 began at 17:58 EDT on 2026-09-28 (seed ledger 107,300,970) and ended with the deployment of 50a05ae at 10:58 EDT, after about 17 hours. It recorded two engine receipts, F407 at 04:46 EDT and F408 at 07:54 EDT, both below. Every check during it found no state-hash mismatch (the last at 09:17 EDT), with the validator on its primary reference node and no failovers: the receipts are differences in the native engine's own results, found by its comparison against libxrpl, not in the validator's state. From 08:26 EDT the comparison itself was out of step (below).
- **Finding F407 (a trust line whose only setting is the default quality).** At ledger 107,311,328 an account sent a TrustSet on a CNY trust line, setting the limit to 0, setting NoRipple and clearing freeze. Afterwards the line held no balance, the account's limit was 0, and NoRipple is that account's default (it does not enable DefaultRipple); the only other thing stored for its side was an incoming quality of 1,000,000,000. That value, QUALITY_ONE, is a rate of exactly 1, the same as no quality at all, and rippled's TrustSet treats a stored QUALITY_ONE as zero before deciding whether either side still has an interest in the line. Neither side did, so mainnet deleted the line: the line and the issuer's directory page went, the account's directory page shrank, and the account's owner count fell from 1 to 0. The engine counted any non-zero stored quality as an interest and kept the line. It now treats a stored QUALITY_ONE as no quality, as rippled does. The mainnet transaction is pinned as a regression vector. Fixed in e1140ef.
- **Finding F408 (AMM deposit: the minimum is not the amount issued).** At ledger 107,314,214 an account deposited 1.021192 XRP into an XRP/token AMM pool as a single-asset deposit, naming 56,639.2869999 LP tokens in the transaction's LPTokenOut field. In a single-asset deposit that field is only a minimum: rippled issues the LP tokens the deposited amount buys at the pool's price, and refuses the deposit (tecAMM_FAILED) only when that is fewer than the minimum. The XRP bought 57,211.40098 tokens, and mainnet issued 57,211.40098. The engine moved the same XRP but issued exactly the minimum, about 572 tokens short. It now issues what the

deposit buys and refuses a deposit that falls below its minimum. The same code decided a second kind of single-asset deposit, in which the account names the tokens it wants: rippled issues that request rounded down to the precision at which the pool's total can record it, and the engine now does too (it issued the unrounded request). The mainnet transaction is pinned as a regression vector; the refusal and the rounding, which no mainnet transaction has exercised yet, are covered by unit tests. Fixed in 1fd1b9b.

- **The native-engine comparison lost step at 08:26 EDT, and was repaired.** Before verifying a ledger with libxrpl, the validator checks that its stored state is aligned with that ledger: the sender of the ledger's first sequence-numbered transaction must hold that sequence number. At ledger 107,314,806 that sender's account flags (it blocks incoming trust lines and NFT offers, flags 0x24000000) begin with the same byte that marks the sequence field, and the check searched for that byte instead of reading the fields in order, so it read 36 where the account held 62,037,183. It concluded the state was out of step, declined to verify the ledger, and handed the native-engine comparison an empty result; the comparison, which resets its in-memory copy of the ledger to the libxrpl result after every ledger, lost that ledger's changes and from then on reported a difference on every ledger: 2,303 ledgers, 107,314,806 to 107,317,138, until the deployment rebuilt its copy. None were engine errors and the validator was unaffected: its state hash matched the network on every ledger throughout. Of the flagged ledgers the automated re-check reached, 65 matched mainnet on every transaction (7,398 of 7,398); the other two, 107,314,819 and 107,314,838, match on the full ledger state with the flow engine the validator runs (105 of 105 and 251 of 251), the re-check having used the engine's older flow code for the first and an incomplete pre-state for one failed payment in the second. Fixed in 50a05ae: the check reads the account's fields in order, and a ledger the libxrpl side did not verify now makes the comparison rebuild its copy from the validator's stored state instead of adopting an empty result. A replay against mainnet of one ledger in twenty from the affected range, with the flow engine the validator runs, matched on every transaction: 117 ledgers, 16,302 of 16,302 transactions (finished 16:17 EDT).
- **Deployment.** Build 50a05ae (F407, F408 and the fix above) replaced the live build at 10:58 EDT, after its gate (1,697 suite tests and 745 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.6 seconds and resumed two ledgers behind the network. The native-engine comparison rebuilt its copy of the ledger (19,997,488 objects in 242 seconds) and has reported no difference since. The health check matched 411 ledgers with no mismatches, and soak #36 began at 11:25 EDT (seed ledger 107,317,138).
- **Soak #36.** At 18:18 EDT, 6.9 hours in, soak #36 of build 50a05ae was healthy: 6,801 state-hash matches and no mismatches, level with the network, on its primary reference node with no failovers, and no engine receipts. The native-engine comparison agreed byte for byte on all 6,795 ledgers it compared, and detected the one-drop canary it planted to prove the comparison is not blind.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Sponsor-paid NFTokenPage reserves for mint-to-player** (nft) This extends Sponsor (XLS-68) to cover `NFTokenPage`, a shared object that holds many tokens and that the current sponsorship model does not handle. A
- **Not-before ledger bound for pre-signed transactions** (hooks) Add a common optional field `sfFirstLedgerSequence` (`UInt32`) that mirrors `LastLedgerSequence`. It is covered by the signature. `preflight` rejects a
- **Directed Offers for RFQ block-trade settlement** (settlement) Add a flag `tfDirected` and the field `sfDestination` to `OfferCreate`. A directed offer never crosses on creation and is never put in a `BookDirector`

Appendix — commits

Time	Repo	Commit	Subject
04:53	xrpl-validator	<code>e1140ef</code>	fix(xrpl-ledger): F407 — a stored QualityIn of QUALITY_ONE is no interest; the default line is deleted
08:16	xrpl-validator	<code>1fd1b9b</code>	fix(xrpl-ledger): F408 — an AMM deposit mints the tokens its mode computes; LPTokenOut is never the mint
09:37	xrpl-validator	<code>50a05ae</code>	fix(xrpl-node): the era sentinel walks the AccountRoot's fields; an unverified ledger re-hydrates the native shadow